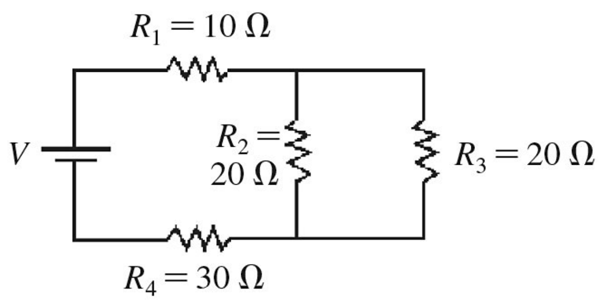
Exam Spring 2026 2054

Name___________________________________

1) Which one of the following types of electromagnetic wave travels through space the fastest? 1) _______

A) ultraviolet
B) microwaves
C) infrared
D) radio waves
E) They all travel through space at the same speed.

2) What is the equivalent resistance in the circuit shown in the figure? 2) _______



A) 55 Ω B) 35 Ω C) 50 Ω D) 80 Ω

3) The maximum value of the electric field in an electromagnetic wave is 2.0 V/m. What is the maximum value of the magnetic field in that wave? ($c = 3.0 \times 10^8$ m/s) 3) _______

A) 6.7 nT B) 6.7 pT C) 6.7 T D) 6.7 μT E) 6.7 mT

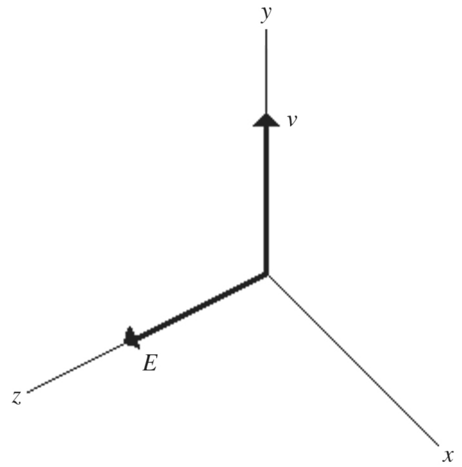
4) When a magnet is moved to and fro in a wire coil, voltage is induced. If the coil has twice as many loops, the voltage induced is 4) _______

A) half.
B) twice.
C) four times as much.
D) the same.
E) none of the above

5) A doubly charged ion with speed 6.9×10^6 m/s enters a uniform 0.80-T magnetic field, traveling perpendicular to the field. Once in the field, it moves in a circular arc of radius 30 cm. What is the mass of this ion? (hint: $e = 1.60 \times 10^{-19}$ C so q = 2e) 5) _______

A) 8.2×10^{-27} kg B) 3.3×10^{-27} kg
C) 11×10^{-27} kg D) 6.7×10^{-27} kg

6) An electromagnetic wave travels in free space in the $+y$ direction, as shown in the figure. If the electric field at the origin is along the $+z$ direction, what is the direction of the magnetic field? 6) _______



A) $-z$ B) $+z$ C) $+y$ D) $+x$ E) $-x$

7) A 12-V battery is connected across a 100-Ω resistor. How many electrons flow through the wire in 1.0 min? ($e = 1.60 \times 10^{-19}$ C and I = q/t) 7) _______

A) 2.5×10^{19} B) 4.5×10^{19} C) 3.5×10^{19} D) 1.5×10^{19}

8) A constant uniform magnetic field of 0.50 T is applied at right angles to the plane of a flat rectangular loop of area 3.0×10^{-3} m^2. If the area of this loop changes steadily from its original value to a new value of 1.6×10^{-3} m^2 in 1.6 s, what is the emf induced in the loop? 8) _______

A) 9.0×10^{-2} V

B) 7.5×10^{-2} V

C) 4.4×10^{-4} V

D) 0 V

E) 1.6×10^{-2} V

9) About 1350 W/m^2 (S) of electromagnetic energy reaches the upper atmosphere of the earth from the sun, which is 1.5×10^{11} m away. Use this information to estimate the average power output of the sun. (P in Watts) 9) _______

A) 4×10^{26} W B) 2×10^{26} W C) 1×10^{26} W D) 3×10^{26} W

10) Which of the following statements about electromagnetic waves in free space are true? (There could be more than one correct choice.) 10) ______

A) The higher-frequency travel faster than the lower-frequency waves.
B) The higher-frequency waves have shorter wavelengths than the lower-frequency waves.
C) The wavelengths of the visible waves are some of the longest electromagnetic waves.
D) The wavelengths of the visible waves are some of the shortest electromagnetic waves.
E) The electric field vector is always at right angles to the magnetic field vector.

11) A simple circuit has a total resistance of 30 Ω. If a 2.0-A current is maintained in this circuit, how much energy is dissipated in this circuit in 4.0 seconds? 11) ______

A) 480 J B) 4.8 J C) 24 J D) 48 J E) 6.0 J

12) A small object with a 5.0-μC charge is accelerating horizontally on a friction-free surface at 0.0050 m/s/s due only to an electric field. If the object has a mass of 2.0 g, what is the magnitude of the electric field?(1 kg = 2000g) 12) ______

A) 0.0020 N/C
B) 1.0 N/C
C) 4.0 N/C
D) 2.0 N/C
E) 0.0040 N/C

13) A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are perpendicular. 13) ______

A) $0\ \mathrm{T \cdot m^2}$ B) $3.1 \times 10^{-3}\ \mathrm{T \cdot m^2}$
C) $5.5 \times 10^{-3}\ \mathrm{T \cdot m^2}$ D) $6.3 \times 10^{-3}\ \mathrm{T \cdot m^2}$

14) Which one of the following lists gives the correct order of the electromagnetic spectrum from *low to high frequencies*? 14) ______

A) radio waves, ultraviolet, x-rays, microwaves, infrared, visible, gamma rays
B) radio waves, microwaves, infrared, visible, ultraviolet, x-rays, gamma rays
C) radio waves, infrared, x-rays, microwaves, ultraviolet, visible, gamma rays
D) radio waves, microwaves, visible, x-rays, infrared, ultraviolet, gamma rays
E) radio waves, infrared, microwaves, ultraviolet, visible, x-rays, gamma rays

15) If an electron is accelerated from rest through a potential difference of 1500 V, what speed does it reach? ($e = 1.60 \times 10^{-19}$ C , $m_{\mathrm{electron}} = 9.11 \times 10^{-31}$ kg) 15) ______

A) 1.9×10^{7} m/s B) 1.1×10^{7} m/s
C) 2.3×10^{7} m/s D) 1.5×10^{7} m/s

16) A step-up transformer steps up voltage by ten times. If voltage input is 120 volts, voltage output is 16) ______

A) 12000 V.
B) 60 V.
C) 120 V.
D) 1200 V.
E) none of the above

17) A straight 1.0-m long wire is carrying a current. The wire is placed perpendicular to a magnetic field of strength 0.20 T. If the wire experiences a force of 0.60 N, what is the current in the wire? 17) ______

A) 1.0 A B) 4.0 A C) 5.0 A D) 3.0 A E) 2.0 A

18) How much time does it take a beam of light to travel 2.9 km through space? ($c = 3.0 \times 10^8$ m/s) 18) ______

A) 9.7 μs B) 9.7 ps C) 9.7 ns D) 9.7 s E) 9.7 ms

19) An electron moves with a speed of 5.0×10^4 m/s perpendicular to a uniform magnetic field of magnitude 0.20 T. What is the magnitude of the magnetic force on the electron? ($e = 1.60 \times 10^{-19}$ C) 19) ______

A) 4.4×10^{-14} N
B) 2.6×10^{-24} N
C) 1.6×10^{-15} N
D) zero
E) 5×10^{-20} N

20) Two point charges each experience a 1-N electrostatic force when they are 2 cm apart. If they are moved to a new separation of 8 cm, what is the magnitude of the electric force on each of them? 20) ______

A) 2 N B) 1/2 N C) 1/4 N D) 1/16 N E) 1/8 N

21)

ELECTRICITY AND MAGNETISM

$$\left|\vec{F}_E\right| = \frac{1}{4\pi\varepsilon_0}\frac{\left|q_1 q_2\right|}{r^2}$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\left|\vec{E}\right| = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2}$$

$$\Delta U_E = q\Delta V$$

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

$$\left|\vec{E}\right| = \left|\frac{\Delta V}{\Delta r}\right|$$

$$\Delta V = \frac{Q}{C}$$

$$C = \kappa\varepsilon_0\frac{A}{d}$$

$$E = \frac{Q}{\varepsilon_0 A}$$

$$U_C = \frac{1}{2}Q\Delta V = \frac{1}{2}C(\Delta V)^2$$

A = area
B = magnetic field
C = capacitance
d = distance
E = electric field
$\mathcal{E}$ = emf
F = force
I = current
ℓ = length
P = power
Q = charge
q = point charge
R = resistance
r = separation
t = time
U = potential (stored) energy
V = electric potential
v = speed
κ = dielectric constant
ρ = resistivity
θ = angle
Φ = flux

$$I = \frac{\Delta Q}{\Delta t}$$

$$R = \frac{\rho\ell}{A}$$

$$P = I\,\Delta V$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$B = \frac{\mu_0}{2\pi}\frac{I}{r}$$

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\left|\vec{F}_M\right| = \left|q\vec{v}\right|\left|\sin\theta\right|\left|\vec{B}\right|$$

$$\vec{F}_M = I\vec{\ell} \times \vec{B}$$

$$\left|\vec{F}_M\right| = \left|I\vec{\ell}\right|\left|\sin\theta\right|\left|\vec{B}\right|$$

$$\Phi_B = \vec{B}\cdot\vec{A}$$

$$\Phi_B = \left|\vec{B}\right|\cos\theta\left|\vec{A}\right|$$

$$\mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t}$$

$$\mathcal{E} = B\ell v$$

Answer Key
Testname: FINAL_106_2024

1) E
2) C
3) A
4) B
5) C
6) D
7) B
8) C
9) A
10) B, E
11) A
12) D
13) D
14) B
15) C
16) D
17) D
18) A
19) C
20) D
21)